

Case Study: Reducing Customer Churn Through Predictive Analytics

Background

Customer churn is one of the most significant challenges faced by subscription-based businesses. When customers leave, organizations lose recurring revenue and must invest additional resources in acquiring new customers.

A telecom company sought to better understand why customers leave and whether machine learning could be used to identify high-risk customers before churn occurs.

The Challenge

The company maintained a large customer database containing demographic information, service subscriptions, billing records, and contract details. However, it lacked a systematic method for identifying customers likely to leave.

Without predictive insights, retention efforts were reactive rather than proactive.

Objective

The objective was to develop a machine learning solution capable of:

- Predicting customer churn.
- Identifying key churn drivers.
- Supporting data-driven retention strategies.

Approach

The project followed a structured analytics workflow.

Data Preparation

Customer data was cleaned and prepared for analysis. Missing values were handled, categorical variables were encoded, and numerical features were standardized where required.

Exploratory Data Analysis

EDA was conducted to understand customer behavior patterns and identify relationships between customer attributes and churn.

Key observations included:

- Customers with shorter tenure exhibited higher churn rates.
- Month-to-month contracts were associated with increased churn.
- Customers using additional support services were more likely to remain with the company.

Model Development

Three machine learning algorithms were trained and evaluated:

- Logistic Regression
- Random Forest
- XGBoost

Performance was measured using Accuracy, Precision, Recall, F1 Score, and ROC-AUC.

Results

Model	Accuracy	ROC-AUC
Logistic Regression	82.04%	0.8620
Random Forest	79.13%	0.8402
XGBoost	81.41%	0.8610

Logistic Regression achieved the strongest overall performance and was selected as the final model.

Business Insights

The model revealed several important business findings:

1. Customer tenure is the strongest retention indicator.
2. Long-term contracts significantly reduce churn.
3. Fiber Optic customers are more likely to leave.
4. Electronic Check users exhibit higher churn rates.
5. Online Security and Tech Support improve customer retention.

Business Impact

The solution enables the company to:

- Identify high-risk customers proactively.
- Improve retention campaign effectiveness.
- Reduce customer acquisition costs.
- Increase customer lifetime value.
- Support data-driven decision-making.

Recommendations

- Introduce loyalty programs for new customers.

- Promote annual and multi-year contract plans.
- Encourage customers to adopt automatic payment methods.
- Bundle support and security services into subscription packages.
- Target high-risk customers with personalized retention offers.

Conclusion

This project demonstrates how machine learning can be leveraged to transform customer data into actionable business intelligence. By accurately predicting churn and identifying its key drivers, organizations can take proactive measures to improve customer retention and protect revenue growth.